

EE 114: PROBABILITY, RANDOM VARIABLES, AND RANDOM PROCESSES

Credits and Contact Hours

4.0 Units

Lecture: MW 1530 – 1650 hours

MRB 103

Discussion Sections: W 2000 – 2050 hours WCH 143

Office Hours:

Instructor: M 1400 – 1500 hours MRB 3128

Instructor and TA

Amit K. Roy-Chowdhury, Professor, Dept. of Electrical and Computer Engineering;
MRB 3128

TA: Sarosij Bose

Textbooks and Related Materials

1. (Text) Roy D. Yates and David J. Goodman, “Probability and Stochastic Processes – A Friendly Introduction for Electrical and Computer Engineers”, 2nd Edition, John Wiley and Sons Inc.
2. (Reference) A. Leon-Garcia, “Probability and Random Processes for Electrical Engineering”, 2nd Edition, Addison Wesley Longman.

Specific Course Information

A. Course Description (Catalog description)

Covers fundamentals of probability theory, random variables, and random processes with applications to electrical and computer engineering. Includes probability theory, random variables, densities, functions of random variables, expectations and moments, and multivariate distributions. Also addresses random processes, autocorrelation function, spectral analysis of random signals, and linear systems with random inputs.

B. Prerequisite(s)

EE 110A, may be taken concurrently or EE 111, may be taken concurrently.

C. Course Type

Electrical Engineering, Computer Engineering, Robotics, required.

D. Grading

Homeworks – 10% (top 5 will be considered)

Mid-terms (2) – Each 20%

Final – 50%

Specific Goals

A. Course Objectives

1. Introduction to basic concepts of probability.
2. Introduction to basic concepts of random variables.
3. Understanding the role of probability and statistics in real life applications.
4. Applications of probability and random variables in electrical and computer engineering.
5. Introduction to multiple random variables, joint distributions, independence and correlation.
6. Introduction to random processes and their applications.
7. Ability to independently solve problems in probability, random variables and random processes.

B. Student Outcomes Addressed

Item	COURSE OBJECTIVES	OUTCOMES										
		A	B	C	D	E	F	G	H	I	J	K
1	Introduction to basic concepts of probability	x										
2	Introduction to basic concepts of random variables	x										
3	Understanding the role of probability and statistics in real life applications	x										
4	Applications of probability and random variables in electrical and computer engineering.					x						
5	Introduction to multiple random variables, joint distributions, independence and correlation	x										
6	Introduction to random processes and their applications	x										
7	Ability to independently solve problems in probability, random variables and random processes	x				x						

- A. Ability to apply knowledge of mathematics, science and engineering
- B. Ability to design and conduct experiments, as well as, analyze and interpret data.
- C. Ability to design a system, component, or process to meet desired needs.
- D. Ability to function on multidisciplinary teams.
- E. Ability to identify, formulate and solve engineering problems.
- F. Understanding of professional and ethical responsibility.
- G. Ability to communicate effectively.
- H. Broad education necessary to understand the impact of engineering solutions in a global and societal context.

- I. Recognition of the need for and an ability to engage in lifelong learning.
- J. Knowledge of contemporary issues.
- K. Ability to use techniques, skills, and modern engineering tools necessary for engineering practice.

List of Topics

- 1) Basic Concepts of Probability Theory
 - Specifying random experiments
 - The axioms of probability
 - Conditional probability
 - Independence of events
 - Applications: Machine Learning (ML) and Signal Processing (SP)
- 2) Random Variables
 - The notion of a random variable
 - The cumulative distribution function
 - The probability density function
 - Some important random variables
 - Functions of a random variable
 - The expected value of random variables
 - Applications: Classification, Noise Models
- 3) Multiple Random Variables
 - Pairs of random variables
 - Independence of two random variables
 - Conditional probability and expectation
 - Random vectors
 - Correlation and covariance
 - Applications: Attention in ML models; Signal Denoising
- 4) Random Processes
 - Definition and properties of a random process
 - Example processes
 - Stationarity
 - Applications of random processing in SP, ML, and Robotics